

Complete Generative AI Roadmap

1. What is Generative AI?

Traditional AI vs Generative AI

Earlier AI systems mainly focused on:

- Classification
- Prediction
- Recommendation

Examples:

- Spam detection
- Movie recommendations
- Face recognition

But Generative AI can:

- Create new content
- Generate human-like responses
- Write code
- Generate images
- Create videos
- Clone voices

How Does Generative AI Work?

Generative AI learns patterns from massive amounts of data.

For example:

If an AI model is trained on billions of sentences, it learns:

- Which word usually comes after another
- How humans communicate
- How context changes meaning
- How to respond naturally

After learning these patterns, it can generate completely new text.

Popular Generative AI Examples

- OpenAI — ChatGPT
- Google — Gemini
- Meta — Llama
- Anthropic — Claude
- Midjourney
- Stability AI — Stable Diffusion

2. Why is Generative AI Such a Big Revolution?

Because it can:

- Produce human-like output
- Understand natural language
- Perform multiple tasks
- Mimic creativity
- Automate complex workflows

This is why many people compare GenAI to:

- The invention of the Internet
- Smartphones
- Cloud computing

3. The Evolution Journey of AI

Understanding the evolution of AI is extremely important.

Phase 1: Traditional Programming

Earlier systems worked like this:

Input + Rules → Output

Example:

if marks > 80:

```
print("A+")
```

The developer manually wrote all the rules.

Phase 2: Machine Learning

Now the machine learns the rules automatically.

Input + Output Data → Machine learns Rules

Examples:

- House price prediction
- Fraud detection
- Customer churn prediction

Phase 3: Deep Learning

Deep Learning uses Neural Networks inspired by the human brain.

Applications:

- Image recognition
- Speech recognition
- Natural Language Processing (NLP)

Phase 4: Transformers (Most Important)

In 2017, Google introduced the Transformer Architecture.

This changed the entire AI industry.

Why Transformers became revolutionary:

- Parallel processing
- Better context understanding
- Long sequence handling
- Scalable training

Modern models built using Transformers:

- GPT

- BERT
- Llama
- Gemini

4. What is a Foundation Model?

This is the core concept of the roadmap.

Definition

A Foundation Model is:

- A massive pre-trained model
- Trained on huge internet-scale data
- General-purpose
- Capable of performing multiple tasks

Example

One single LLM can:

- Translate languages
- Write code
- Summarize documents
- Answer questions
- Reason logically

All using the same model.

5. What is an LLM (Large Language Model)?

LLMs are text-based foundation models.

Examples:

- GPT-4
- Claude
- Llama
- Gemini

What Does an LLM Learn?

LLMs learn:

- Grammar
- Facts
- Human conversation style
- Programming syntax
- Reasoning patterns

6. What is an LMM (Large Multimodal Model)?

LMMs can process:

- Text
- Images
- Audio
- Video

Example:

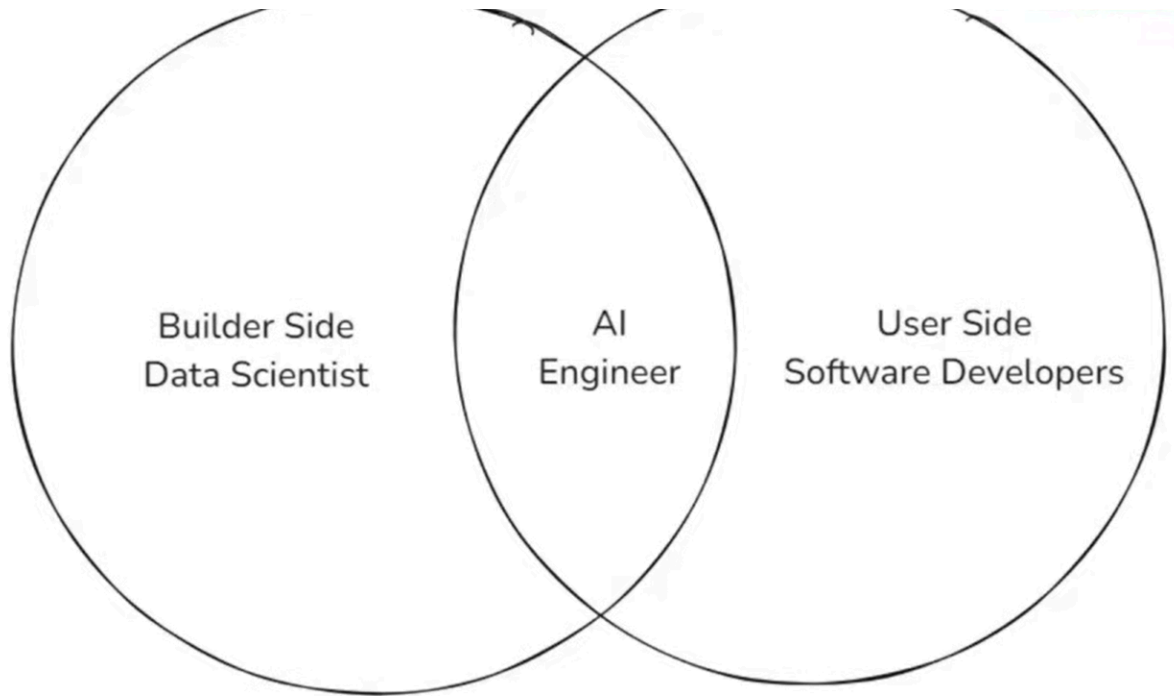
You upload an image and ask:

“What is happening in this image?”

The model can analyze and answer.

7. The Two Main Tracks of the Roadmap

This is the most important section of the roadmap.



TRACK 1 — AI Application Engineer (User Side)

This is the most job-oriented path.

In this track, you:

- Use existing AI models
- Build AI applications
- Create AI-powered products

Currently, this is one of the highest-demand skill sets.

STEP-BY-STEP USER SIDE ROADMAP

STEP 1: Programming Foundation

Learn Python First

The entire AI ecosystem is heavily Python-based.

Topics to learn:

- Variables
- Loops
- Functions
- OOP (Object-Oriented Programming)
- File handling

Important Libraries

- NumPy
- Pandas
- Matplotlib

These are essential for data handling and visualization.

STEP 2: Understanding APIs

You will learn how to use AI models through APIs.

Examples:

- ChatGPT API
- Gemini API
- Claude API

Important Concepts

Learn:

- REST APIs
- JSON
- Authentication
- API keys

Practical Projects

Build:

- AI chatbot

- AI email writer
- AI summarizer

STEP 3: Prompt Engineering

This is one of the most underrated skills in GenAI.

The same model can produce:

- Poor output with bad prompts
- Excellent output with good prompts

Types of Prompting

Zero-shot Prompting

Direct instruction.

Example:

Write a blog about AI.

Few-shot Prompting

Giving examples before asking the task.

Chain-of-Thought Prompting

Asking the model to think step-by-step.

Role Prompting

Example:

Act as a doctor.

STEP 4: RAG (Retrieval-Augmented Generation)

RAG is extremely important.

Why?

Because LLM knowledge may become outdated.

RAG allows models to:

- Search external documents
- Retrieve relevant information
- Generate accurate answers

Example

Building a chatbot using company PDFs.

When the user asks a question:

- The system searches the PDFs
- Retrieves relevant information
- Generates an answer

Components of RAG

Embedding Model

Converts text into vectors.

Vector Database

Examples:

- Pinecone
- Weaviate
- ChromaDB
- FAISS

Retriever

Fetches relevant documents.

Generator (LLM)

Generates the final answer.

STEP 5: Fine-Tuning

Fine-tuning means training a model using your own custom data.

Use Cases

- Medical AI assistant
- Legal chatbot
- Bengali language assistant

Types of Fine-Tuning

Full Fine-Tuning

Very expensive.

PEFT (Parameter-Efficient Fine-Tuning)

More practical and popular.

Examples:

- LoRA
- QLoRA

STEP 6: AI Agents

AI Agents are considered the future of AI.

Agents do not just answer —
they take actions.

Example

You say:

“Book a ticket from Dhaka to Cox’s Bazar.”

The AI Agent can:

- Open a browser
- Search websites
- Fill forms
- Proceed to payment pages

Important Agent Topics

- Tool calling
- Function calling
- Planning
- Memory systems
- Multi-agent collaboration

Popular Frameworks

- LangChain
- LangGraph
- CrewAI
- AutoGen

STEP 7: Deployment & LLMOps

This is where you learn production-level AI engineering.

Topics

Docker

Containerization technology.

FastAPI

Building AI backend APIs.

Cloud Deployment

Learn:

- AWS
- GCP
- Azure

Monitoring

Track:

- Latency
- Hallucinations
- API costs
- Performance

TRACK 2 — AI Researcher / Model Builder

This path is much more advanced and technical.

STEP 1: Strong Mathematics

Math is extremely important here.

Topics

Linear Algebra

Learn:

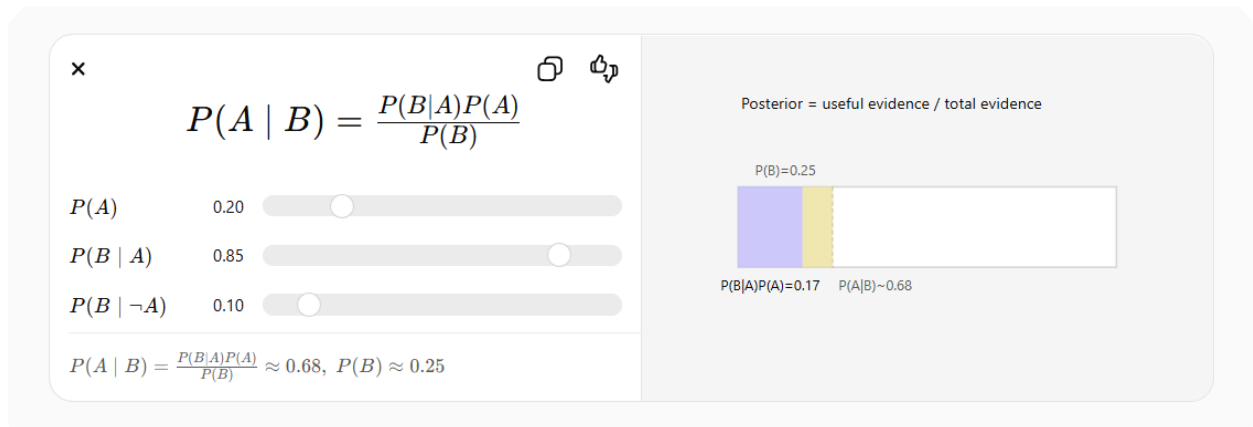
- Vectors
- Matrices
- Eigenvalues

Probability

Learn:

- Bayes Theorem
- Probability distributions

One of the most important formulas:



Calculus

Learn:

- Gradients
- Derivatives
- Optimization

STEP 2: Machine Learning

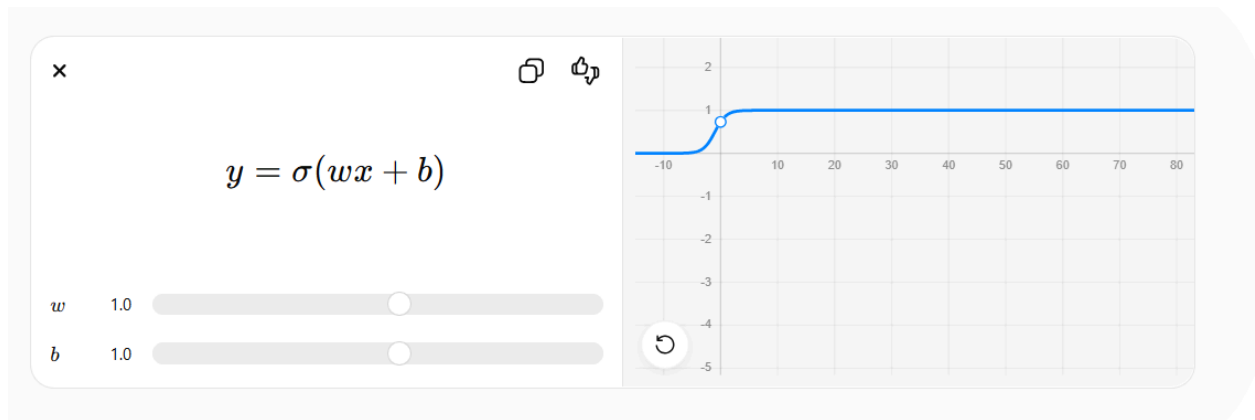
Core ML topics:

- Regression
- Classification
- Clustering

STEP 3: Deep Learning

The most important concept:

Neural Networks



CNNs

Used for image-related tasks.

RNNs / LSTMs

Used for sequence modeling.

STEP 4: PyTorch

PyTorch is the industry-standard deep learning framework.

Learn:

- Tensors
- Autograd
- Training loops
- GPU training

STEP 5: Transformers Deep Dive

Transformers are the heart of GenAI.

Key Concepts

Attention Mechanism

The most important transformer formula:

$$\mathrm{Attention}(Q,K,V)=\mathrm{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$

Self-Attention

Allows the model to understand context relationships.

Encoder & Decoder

- Encoder → Understands input
- Decoder → Generates output

STEP 6: Tokenization

Text is converted into tokens.

Example:

"I love AI"

Becomes:

["I", "love", "AI"]

STEP 7: Pretraining

Models are trained on huge datasets.

Training costs may involve:

- Millions of dollars
- Thousands of GPUs

STEP 8: RLHF

RLHF = Reinforcement Learning from Human Feedback.

This technique helped make ChatGPT more helpful and aligned.

STEP 9: Model Optimization

Quantization

Reducing model size.

Distillation

Creating smaller versions of large models.

Pruning

Removing unnecessary weights.

8. Essential Tools & Technologies

Beginner Tools

- Python
- Jupyter Notebook
- VS Code
- Git/GitHub

Intermediate Tools

- Hugging Face
- LangChain
- Ollama

Advanced Tools

- Kubernetes
- Ray
- vLLM
- TensorRT

9. Best Portfolio Projects

Beginner Projects

- AI chatbot
- PDF Q&A bot
- AI resume analyzer

Intermediate Projects

- RAG chatbot
- AI coding assistant
- AI tutor

Advanced Projects

- Multi-agent systems
- Voice assistant
- Autonomous research agent

10. High-Demand Job Roles in GenAI

Currently popular roles:

- AI Engineer
- LLM Engineer
- Prompt Engineer
- AI Product Developer
- ML Engineer
- Applied AI Researcher

11. Most Important Advice

Don't Learn Only Theory

Build projects every day.

Projects Matter More Than Certificates

Companies want to see:

“What can you build?”

Contribute to Open Source

Especially:

- Hugging Face
- LangChain
- Ollama ecosystem

13. Recommended Learning Resources

YouTube Channels

- CampusX
- Andrej Karpathy
- 3Blue1Brown

Platforms

- [Coursera](#)
- [DeepLearning.AI](#)
- [Hugging Face Course](#)

Documentation

- [PyTorch Docs](#)
- [LangChain Docs](#)
- [OpenAI Docs](#)